

AI for Economic Research: Dynamic Models, Language, and Agents

Lecture 6.5a: Mean-Field Games — Motivation and HJB

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Topic 6.5a Agenda

1. **Mean-Field Games** — motivation, history, two-equation system
2. **HJB side** — continuous-time control, Poisson income shocks

*Arc: T1 Aiyagari + computation → T2a Why ML / Euler → T2b Global DNN → T3 KS / DeepHAM → T4 DEQN / OLG → **T5a MFG HJB** → T5b MFG KFE + Lab*

Mean-Field Games

Mean-Field Games: motivation

- All the models we've seen so far are *finite or large-but-finite* populations of agents (Aiyagari has a continuum but discretises numerically).
- **Mean-Field Games** (MFG) provide a continuous-time, continuum-of-agents formulation.
- Why care?
 - Continuous time often *simplifies* the math: PDEs replace functional equations.
 - Powerful new toolbox for HA models with idiosyncratic and aggregate shocks.
 - Tight link to optimal transport, stochastic analysis, and modern PDE theory.
 - Recent breakthroughs in *deep learning for MFG* (Carmona–Laurière, Achdou–Lasry–Lions, Han et al.).
- This part is an introduction; you can think of MFG as the continuum limit of Aiyagari/KS.

The core idea: continuum + mean-field feedback

- Imagine a *continuum* of identical agents, each solving an individual control problem.
- Each agent's payoff depends on the *distribution* of all other agents (the “mean field”).
- The distribution is itself determined by the optimal behaviour of the agents.
- This circular dependence is the defining feature of MFG — and is exactly the same fixed point as in HA models, but stated in continuous time.
- Two mathematical objects describe the equilibrium:
 - A backward Hamilton–Jacobi–Bellman (HJB) equation for the individual value function.
 - A forward Kolmogorov forward / Fokker–Planck (KFE) equation for the distribution.

A brief history

- **Engineering origins** (mid-2000s):

Huang, Malhamé & Caines (2006, *Communications in Information and Systems* 6(3):221–252) developed “Nash certainty equivalence” for large-population games.

- **Mathematical formalisation:**

Lasry & Lions (2007, *Japanese Journal of Mathematics*) coined the term “Mean-Field Games” and developed the rigorous PDE theory.

- **Economic adoption:**

Achdou–Han–Lasry–Lions–Moll (2014, 2022 *REStud*) translated the framework into HA macroeconomics — continuous-time Aiyagari–Bewley–Huggett.

- **Deep-learning era** (2020s):

Carmona–Laurière, Han–Hu, Achdou–Pequito, . . . use deep neural networks to solve MFGs in high dimensions.

The two-equation MFG system

Hamilton–Jacobi–Bellman (backward, individual):

$$-\partial_t v(x, t) + \rho v(x, t) = H(x, \nabla v(x, t), m(\cdot, t)),$$

where H is the Hamiltonian of the individual control problem and $m(\cdot, t)$ is the population distribution at time t .

Kolmogorov forward / Fokker–Planck (forward, aggregate):

$$\partial_t m(x, t) = -\nabla \cdot (s^*(x, t, m) m(x, t)) + \frac{1}{2} \Delta (\sigma^2(x, t) m(x, t)),$$

where s^* is the optimal drift derived from the HJB.

- HJB and KFE are *coupled*: the optimal control depends on m , and m evolves under the optimal control.
- In equilibrium, both equations must hold simultaneously.

Why HJB is familiar, why KFE is new

- **HJB** is the continuous-time limit of the Bellman equation. Macroeconomists have seen it many times: Merton's portfolio choice, deterministic optimal growth, ...
- **KFE / Fokker–Planck** describes the time evolution of a probability density under stochastic dynamics.
 - Originated in physics: Fokker (1914), Planck (1917) — diffusion of particles.
 - Formalised by Kolmogorov (1931).
 - Standard in physics and engineering, only recently mainstream in macro.
- Together they describe the *joint dynamics of optimal individual behaviour and the cross-sectional distribution* — exactly what HA models need.

HJB review: deterministic optimal control

A generic optimal control problem:

$$v(x_0) = \max_{\alpha(\cdot)} \int_0^\infty e^{-\rho t} r(x_t, \alpha_t) dt \quad \text{s.t.} \quad \dot{x}_t = f(x_t, \alpha_t), \quad x_0 \text{ given.}$$

The corresponding HJB equation:

$$\rho v(x) = \max_{\alpha} \{r(x, \alpha) + v'(x) f(x, \alpha)\}.$$

Interpretation:

- LHS: “required return” on the value function.
- RHS: instantaneous payoff plus the marginal value of the state moving in the chosen direction.
- First-order condition over α : $r_{\alpha} + v'(x)f_{\alpha} = 0$.

From discrete-time Bellman to continuous-time HJB

Start with the discrete-time Bellman over a small interval Δ :

$$v(x) \approx \max_{\alpha} \{ r(x, \alpha) \Delta + e^{-\rho \Delta} v(x + f(x, \alpha) \Delta) \}.$$

Subtract $v(x)$ from both sides, divide by Δ , and let $\Delta \rightarrow 0$ (using $e^{-\rho \Delta} \approx 1 - \rho \Delta$ and

$v(x + f \Delta) \approx v(x) + v'(x) f \Delta$):

$$0 = \max_{\alpha} \{ r(x, \alpha) - \rho v(x) + v'(x) f(x, \alpha) \}.$$

Rearrange:

$$\rho v(x) = \max_{\alpha} \{ r(x, \alpha) + v'(x) f(x, \alpha) \}.$$

This is exactly the HJB.

Stationary HJB and FOCs

For the optimal growth problem with state k , control c , return $u(c)$, and transition $\dot{k} = f(k) - \delta k - c$:

$$\rho v(k) = \max_c \{ u(c) + v'(k) (f(k) - \delta k - c) \}.$$

First-order condition:

$$u'(c) = v'(k) \implies c^*(k) = (u')^{-1}(v'(k)).$$

Substituting back:

$$\rho v(k) = u(c^*(k)) + v'(k) (f(k) - \delta k - c^*(k)).$$

A second-order ODE in v to be solved on $k \in [0, \bar{k}]$ with appropriate boundary conditions.

Poisson income shocks: state space

- In the continuous-time Aiyagari, individual income y jumps between two states $\{y_1, y_2\}$ at *Poisson rates* λ_1, λ_2 .
- State of an individual: (a, y_i) with $i \in \{1, 2\}$.
- Two value functions $v_1(a)$ and $v_2(a)$, one per income state.
- Asset evolves continuously: $\dot{a} = y_i + ra - c$.
- Income switches at Poisson rate; from state i , the agent moves to state $-i$ at rate λ_i .
- This setup is the natural continuous-time analogue of the discrete-time Aiyagari with Markov income.

Deriving the HJB with jump terms (advanced — skip on first pass)

Over a short interval Δ , with probability $\lambda_i \Delta$ the agent's income jumps from y_i to y_{-i} . The Bellman is

$$v_i(a) \approx \max_c \{ u(c) \Delta + e^{-\rho \Delta} [(1 - \lambda_i \Delta) v_i(a + \dot{a} \Delta) + \lambda_i \Delta v_{-i}(a)] \}.$$

Letting $\Delta \rightarrow 0$ and rearranging:

$$\rho v_i(a) = \max_c \{ u(c) + v'_i(a) (y_i + ra - c) + \lambda_i (v_{-i}(a) - v_i(a)) \}.$$

- Three terms: instantaneous utility, drift term (deterministic capital accumulation), jump term (Poisson income switch).
- This is the HJB with jumps; the system is $i \in \{1, 2\}$.

Economic intuition of the jump term

$$\lambda_i(v_{-i}(a) - v_i(a))$$

- This is the *option value* of switching income state.
- If $v_{-i}(a) > v_i(a)$ (the other income state would yield higher continuation value at the same wealth), the jump term is *positive* — the agent enjoys an expected gain from the future income switch.
- Conversely, if $v_{-i}(a) < v_i(a)$, the jump term is a cost.
- The Poisson rate λ_i controls the strength of this option value.
- Intuition: precautionary saving in continuous time appears partly through this jump term, in addition to the curvature of u along the deterministic drift.

Continuous-time Aiyagari: setup

Households. A continuum of households indexed by current asset $a \in [\underline{a}, \infty)$ and income $y_i \in \{y_1, y_2\}$ with Poisson switching rates λ_1, λ_2 .

Firm. Representative firm with $Y = K^\alpha L^{1-\alpha}$ and competitive factor markets:

$$r = \alpha(K/L)^{\alpha-1} - \delta, \quad w = (1 - \alpha)(K/L)^\alpha.$$

Aggregation. Aggregate capital and labour:

$$K = \sum_{i=1}^2 \int a g_i(a) da, \quad L = \sum_{i=1}^2 y_i \int g_i(a) da,$$

where $g_i(a)$ is the cross-sectional density in income state i .

The aggregation loop

1. Guess prices (r, w) (or equivalently aggregate K).
2. Solve the household HJB to get optimal consumption $c_i^*(a)$ and drift $s_i^*(a) = y_i + ra - c_i^*(a)$.
3. Solve the KFE for the stationary density $g_i^*(a)$.
4. Compute aggregate $K^* = \sum_i \int a g_i^*(a) da$.
5. Update (r, w) from the firm FOCs at K^* , and iterate until $K^* = K$.

This is structurally identical to the discrete-time Aiyagari nested fixed point, but every step is now an ODE/PDE rather than a discrete-time DP problem.

Next (T5b): the KFE side — forward equation for the density, upwind discretisation, and the lab preview.